Functional Equations — problem 11.38

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(a) The function $f(x)$ is defined for all $x > 0$ and satisfies the conditions
(1) $f(x)$ is strictly increasing on $(0, +\infty)$,
(2) $f(x) > -1/x$ for $x > 0$,
(3) $f(x) \cdot f(f(x) + 1/x) = 1$ for all $x > 0$.
Find $f(1)$.
(b) Give an example of a function $f(x)$ which satisfies (a).